



Modeling and Forecasting Nepal's Nominal GDP with An Updated Box Jenkins ARIMA Analysis

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ABSTRACT

Background: Gross domestic product (GDP) is one of the principal indicators used to summarize the scale and evolution of economic activity and provides an important information base for macroeconomic planning, fiscal management, investment decisions, and the allocation of public resources. Reliable GDP forecasts are particularly valuable for economies experiencing structural transformation and recurrent external shocks. This study focuses the ARIMA analysis of the annual nominal GDP from 1990/91 to 2024/25. The study aims to model and forecast Nepal's annual nominal GDP using the univariate Box–Jenkins time-series methodology.

Method: The study uses 35 annual observations of Nepal's nominal GDP at current prices, expressed in million Nepalese rupees. The univariate Box Jenkins approach was applied in R to identify and estimate appropriate autoregressive integrated moving average (ARIMA) models. The GDP series was logarithmically transformed to reduce scale effects and stabilize variation. Stationarity was examined using graphical analysis, autocorrelation and partial autocorrelation correlograms, the Augmented Dickey–Fuller (ADF) test, and the Kwiatkowski–Phillips–Schmidt–Shin (KPSS) test. Six pre-specified ARIMA models were estimated using maximum likelihood and compared using Akaike information criterion (AIC), corrected Akaike information criterion (AICc), Bayesian information criterion (BIC), log-likelihood, root mean square error (RMSE), mean absolute deviation (MAD), and residual Ljung–Box diagnostics. An automated model-selection procedure was also used as a robustness check.

Results: The GDP series exhibited a strong non-stationary trend in its original form. After logarithmic transformation and first differencing, the KPSS test supported stationarity, although the ADF test continued to provide mixed evidence regarding the presence of a unit root. Among the six pre specified candidate models, ARIMA(1, 1, 2) produced the lowest AIC (-98.7618) and AICc (-97.3825). The residual Ljung Box test was statistically insignificant at the 5% level, indicating no strong evidence of remaining autocorrelation in the residuals. However, the autoregressive coefficient was estimated at the boundary of the parameter space, resulting in an unavailable standard error and suggesting a potential near unit root estimation issue. The automated model selection procedure identified the more parsimonious ARIMA(0, 1, 0) model with drift and reported lower information criteria than the restricted candidate model. Forecasts based on ARIMA(1, 1, 2) indicate a continued increase in Nepal's nominal GDP from NPR 6,353,083 million in 2025/26 to NPR 7,907,673 million in 2029/30, although the prediction intervals widen substantially as the forecast horizon increases.

Conclusion: The findings demonstrate both the usefulness and limitations of univariate ARIMA modeling for forecasting Nepal's aggregate GDP. Within the restricted set of pre-specified models, ARIMA(1, 1, 2) was identified as the preferred model according to AIC and AICc, while the automated analysis suggested that the simpler ARIMA(0, 1, 0) model with drift provides an important parsimonious benchmark. The results indicate that Nepal's nominal GDP is expected to continue increasing over the forecast period, although forecast uncertainty increases over time. The study highlights the importance of comparing alternative specifications and suggests that future research should evaluate multivariate, hybrid, and other forecasting approaches alongside univariate ARIMA models to improve the robustness and accuracy of GDP forecasts for Nepal.

Key words: ARIMA; Box–Jenkins methodology; GDP forecasting; Nepal; Time series analysis; Nominal GDP.

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INTRODUCTION

Gross domestic product (GDP) is a central summary measure of economic activity. In nominal terms, GDP represents the monetary value of final goods and services produced within an economy at current prices. Changes in nominal GDP can reflect changes in the volume of production, changes in prices, or both. Consequently, nominal GDP and real GDP answer different analytical questions. Real GDP is generally more appropriate when the objective is to study changes in the volume of output, whereas nominal GDP remains highly relevant for fis-

cal aggregates, current-price national accounts, debt and revenue ratios, public expenditure planning, and other decisions formulated in monetary terms.

GDP can be compiled through the production, expenditure, and income approaches. Under the production approach, value added is aggregated across institutional sectors and industries, with the appropriate treatment of taxes and subsidies on products. Under the expenditure approach, GDP is associated with final consumption, gross capital formation, and net exports. Under the income approach, it corresponds to the income gener-

ated through production, including compensation of employees, taxes less subsidies on production and imports, and operating surplus or mixed income. Although these approaches are conceptually related, official national accounts are compiled within an accounting framework designed to provide internally consistent macroeconomic aggregates.

For Nepal, reliable information about the expected trajectory of GDP is important because public sector planning, infrastructure expenditure, fiscal policy, development programmes, and private investment decisions all depend partly on expectations concerning the future size of the economy. Forecasts cannot eliminate uncertainty, particularly when an economy is exposed to policy changes, natural disasters, international price movements, migration and remittance dynamics, or other structural shocks. Nevertheless, transparent statistical forecasts provide a useful baseline against which alternative scenarios and policy assumptions can be assessed.

The Box–Jenkins framework remains a useful starting point for such an exercise because it provides a systematic sequence for identifying non-stationarity, transforming the series where appropriate, examining autocorrelation structures, estimating competing ARIMA specifications, diagnosing residuals, and generating forecasts (Box and Jenkins, 1976; Box et al., 2015; Hyndman et al., 2025). ARIMA models are particularly attractive when the primary objective is to construct a transparent univariate benchmark using the historical dynamics of the series itself.

Recent forecasting research also shows why a classical ARIMA benchmark remains relevant even when more complex methods are available. Nepal-specific work by Chaudhary and Uprety (2023) compares ARIMA, artificial neural networks, and hybrid ARIMA–ANN specifications for GDP forecasting and reports improved predictive performance for ANN and hybrid approaches. Similarly, recent international work continues to compare conventional ARIMA specifications with machine-learning and hybrid methods rather than assuming that a single class of models is universally superior (Atif, 2025; Sinu et al., 2024). These developments again suggest that a carefully diagnosed ARIMA model should be interpreted as an important baseline for macroeconomic forecasting.

The objectives of this study are therefore fourfold:

1. to describe the historical evolution of Nepal's annual nominal GDP from 1990/91 to 2024/25;
2. to investigate the time-series properties of the GDP series using graphical, ADF, and KPSS evidence;
3. to estimate and compare a set of candidate ARIMA models using likelihood-based information criteria, in-sample error measures, and residual diagnostics; and
4. to generate five-year forecasts for 2025/26–2029/30 and interpret the associated forecast uncertainty.

A further contribution of the updated study is methodological transparency. The candidate-model comparison

and the automated `auto.arima()` result do not point to exactly the same preferred specification. Instead of suppressing this difference, the study reports both results and discusses the implications for parsimony and model uncertainty.

REVIEW OF RELATED LITERATURE

ARIMA and the Box–Jenkins tradition

The autoregressive integrated moving-average model is one of the most widely used approaches to univariate time-series modeling. The classical Box–Jenkins framework emphasizes an iterative cycle of identification, estimation, diagnostic checking, and forecasting (Box and Jenkins, 1976; Box et al., 2015). The autoregressive component represents dependence on previous values of the series, the integrated component represents differencing used to address non-stationarity, and the moving-average component represents dependence on previous innovations.

A major advantage of the framework is that model choice is not based solely on one numerical criterion. A useful model should balance goodness of fit with parsimony and should leave residuals that are approximately consistent with a white-noise process. Information criteria such as AIC and BIC provide systematic penalties for unnecessary complexity, while residual diagnostics help determine whether meaningful serial dependence remains after estimation.

More recent Nepal-focused research has expanded the methodological comparison. Upadhyay and Pradhan Upadhyay and Pradhan (2023) applied the autoregressive integrated moving average (ARIMA) methodology to forecast Nepal's GDP, providing an earlier Nepal-specific benchmark for univariate GDP forecasting. Chand (2024) applied ARIMA modeling to Nepal's per-capita GDP using annual data from 1960 to 2022 and identified an ARIMA(2,2,1) specification for long-horizon prediction. The study demonstrates that even closely related macroeconomic indicators may require different orders of integration and different dynamic structures.

A particularly relevant recent contribution is the Nepal Rastra Bank study by Chaudhary and Uprety (2023). The authors examine GDP forecasting using standalone ARIMA and ANN models as well as additive and multiplicative hybrid ARIMA–ANN models. Their findings emphasize that Nepal's GDP dynamics may contain both linear and nonlinear components and that hybrid or neural-network approaches can improve predictive accuracy in some settings. This result does not make ARIMA obsolete; instead, it reinforces the role of ARIMA as a transparent benchmark against which more flexible approaches can be compared.

Recent empirical work has also combined ARIMA forecasting with econometric approaches in examining Nepal's macroeconomic conditions. Pradhan et al. Pradhan et al. (2025) investigated the relationship between foreign direct investment and economic growth in Nepal using ARDL and ARIMA forecasting models. The study illustrates the value of combining explanatory econometric models with time-series forecasting approaches when

analyzing Nepal's economic dynamics.

ARIMA-based forecasting has also been applied to other important macroeconomic and economic indicators in Nepal. Pradhan and Koirala [Pradhan and Koirala \(2024\)](#) analyzed and forecast international tourist arrivals in Nepal, demonstrating the broader applicability of time-series forecasting techniques to Nepalese economic planning. Such applications highlight the continued relevance of statistical forecasting methods for economic indicators beyond GDP. The application of statistical forecasting in Nepal extends beyond macroeconomic aggregates to tourism and agricultural-related indicators. Pradhan and Koirala [Pradhan and Koirala \(2024\)](#) examined trends and forecast international tourist arrivals in Nepal, while Poudel et al. [Poudel et al. \(2025\)](#) applied statistical forecasting methods to cereal grain requirements for livestock and fisheries feed. These studies demonstrate the wider use of time-series forecasting for planning and resource allocation in Nepalese contexts.

GDP forecasting studies across countries: China, India, Greece, Sudan, Bangladesh, Kenya, and Pakistan also report considerable variation in preferred ARIMA specifications. ([Dritsaki, 2015](#); [Hassan, 2020](#); [Maity and Chatterjee, 2012](#); [Miah et al., 2019](#); [Ning et al., 2010](#); [Wabomba et al., 2016](#); [Zakai, 2014](#)). The diversity of reported models supports the principle that ARIMA order should be determined empirically rather than selected by analogy alone.

A recent multi country study further illustrates the continuing use of ARIMA as a baseline in comparisons involving regime switching and deep learning approaches ([Atif, 2025](#)). The broader methodological lesson is that additional complexity should be justified by improved predictive performance and robust diagnostics rather than assumed to be superior a priori.

Taken together, the literature provides three motivations for the present analysis. First, ARIMA remains a useful and interpretable benchmark for macroeconomic forecasting. Second, updated data can materially change the preferred model. Third, comparison with parsimonious alternatives and automated selection procedures is important, especially when a more complex model produces boundary estimates or other numerical warnings.

DATA AND METHODOLOGY

Data source and study period

The study uses annual nominal GDP data for Nepal covering fiscal years 1990/91 to 2024/25, yielding 35 observations. GDP is measured at current prices and expressed in million Nepalese rupees. The updated dataset used in the empirical analysis is stored in the study's online data repository and was read directly into R for cleaning and time-series construction. The analysis should be interpreted with the understanding that official national-account estimates may be revised as statistical methodologies and source data are updated. The study therefore documents the exact sample used for estimation rather than combining multiple data vintages.

For time-series modeling, the fiscal-year label was converted to the first calendar year of the fiscal period. Thus,

1990/91 was indexed as 1990, 1991/92 as 1991, and so forth. The annual frequency is one observation per year:

$$GDP_t, \quad t = 1990, 1991, \dots, 2024. \quad (1)$$

Because nominal GDP rises strongly over the sample and its scale changes substantially over time, the principal ARIMA analysis uses the natural logarithm:

$$y_t = \ln(GDP_t). \quad (2)$$

The logarithmic transformation does not remove every possible source of non-stationarity, but it places the series on a scale in which changes can be interpreted approximately as proportional changes and can reduce the influence of level dependent variance.

The ARIMA model

Let y_t denote a time series. An ARIMA(p, d, q) model can be written as

$$\phi(B)(1 - B)^d y_t = c + \theta(B)\varepsilon_t, \quad (3)$$

where B is the backshift operator, $(1 - B)^d$ denotes differencing of order d , $\phi(B)$ is the autoregressive polynomial, $\theta(B)$ is the moving-average polynomial, and ε_t is an innovation term.

For example, after one difference, the ARIMA(1, 1, 2) model considered in this study can be represented conceptually as

$$(1 - \phi_1 B)(1 - B)y_t = (1 + \theta_1 B + \theta_2 B^2)\varepsilon_t. \quad (4)$$

The exact sign convention for MA parameters depends on the software parameterization, so the fitted coefficients are interpreted according to the implementation used in R.

The Box-Jenkins procedure applied in this study contains four broad stages:

1. **Identification:** inspect the series and assess stationarity using plots, ACF/PACF patterns, and formal unit-root or stationarity tests;
2. **Estimation:** estimate plausible ARIMA specifications by maximum likelihood;
3. **Diagnostic checking:** examine residual behavior using plots, ACF/PACF, normality testing, and Ljung-Box statistics;
4. **Forecasting:** generate out-of-sample forecasts and prediction intervals from the selected specification.

Stationarity tests

Two complementary tests were used. The ADF test has the null hypothesis that the series contains a unit root. Failure to reject the null is therefore interpreted as evidence consistent with non-stationarity. The KPSS test reverses the hypotheses: its null is stationarity around the specified deterministic component, and rejection suggests non-stationarity ([Dickey and Fuller, 1979](#); [Kwiatkowski et al., 1992](#)).

Using both tests is useful because the tests are based on opposite null hypotheses. Nevertheless, their results

do not always agree perfectly in small samples or in the presence of structural changes. Accordingly, the present study interprets the ADF and KPSS evidence jointly rather than treating either test as mechanically decisive.

Candidate model estimation and comparison

The following six ARIMA models were estimated on the log-transformed series:

(1, 1, 1), (1, 1, 2), (2, 1, 1), (2, 1, 2), (6, 1, 1), (6, 1, 2).

Models were estimated using maximum likelihood. Their performance was compared using:

- Akaike information criterion (AIC);
- corrected Akaike information criterion (AICc);
- Bayesian information criterion (BIC);
- log-likelihood;
- root mean squared error (RMSE);
- mean absolute deviation (MAD); and
- Ljung–Box residual test statistics and p -values.

The information criteria balance fit against model complexity, while RMSE and MAD summarize in-sample residual magnitude. The Ljung–Box test evaluates whether residual autocorrelation remains after accounting for the fitted model. A model is not selected solely because it minimizes one error measure; model adequacy also requires reasonable residual behavior and parameter estimation.

Automated model-selection robustness check

As a robustness exercise, the study also applies `auto.arima()` with exhaustive search settings. This procedure searches over a broader set of admissible specifications according to an information criterion. The automated result is not identical to the best model in the restricted candidate set. Reporting this difference is important because it reveals whether the manually specified candidates omitted a simpler or statistically preferred alternative.

Forecasting and uncertainty

Five annual forecasts are generated for 2025/26–2029/30. The study reports point forecasts together with 80% and 90% prediction intervals. Prediction intervals are particularly important because the uncertainty surrounding a time-series forecast generally increases as the horizon extends. The forecasts should therefore be interpreted as conditional statistical projections based on the historical dynamics and model assumptions, not as guaranteed future outcomes.

RESULTS AND DISCUSSION

Historical pattern of Nepal's nominal GDP

The annual GDP series displays a pronounced long-run upward movement over the study period. This pattern is expected for nominal GDP because both real economic

expansion and changes in the general price level contribute to increases in current-price output. The visual trend nevertheless indicates that direct modeling of the untransformed level series with a stationary ARMA process would be inappropriate without further treatment.

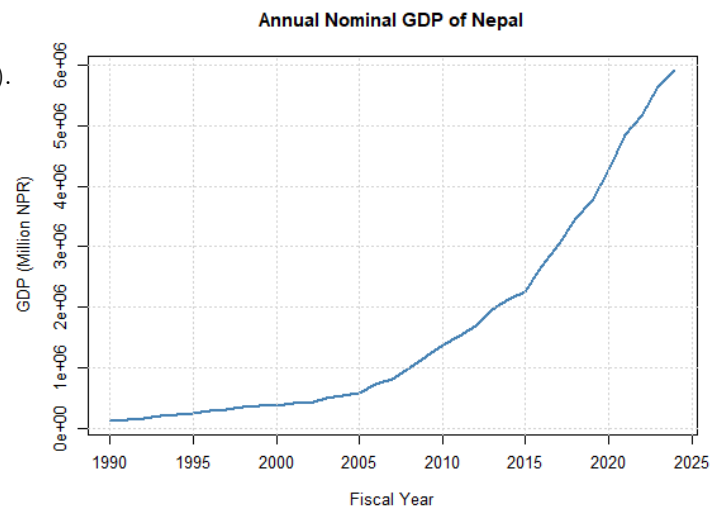


Figure 1. Annual nominal GDP of Nepal, 1990/91–2024/25.

The figure should be discussed in relation to both scale and trend. A steadily rising nominal series does not by itself imply a stable growth process because the increments can vary substantially over time. For this reason, the analysis proceeds with logarithmic transformation and differencing rather than assuming that the visible upward path represents a deterministic trend that can simply be extrapolated.

Stationarity analysis

Table 1 presents the formal stationarity results obtained from the R analysis.

The evidence for the GDP level is consistent across the two tests. The ADF statistic has a very high reported p -value, so the unit-root null is not rejected. Conversely, the KPSS test rejects the null of level stationarity. These findings support the conclusion that the level series is non-stationary.

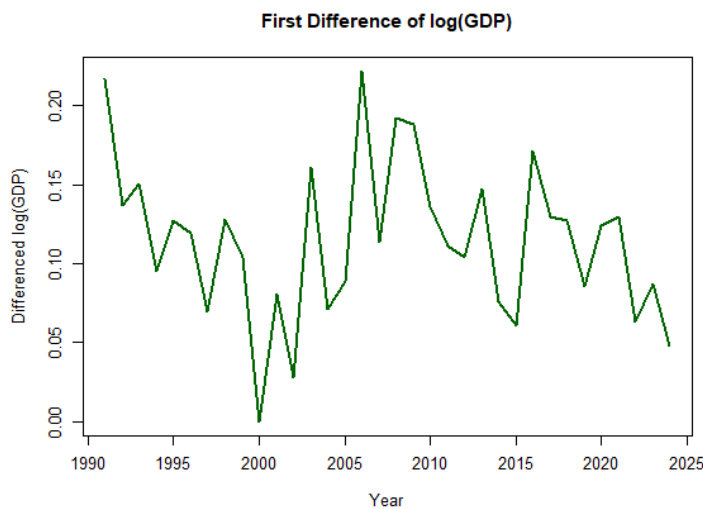
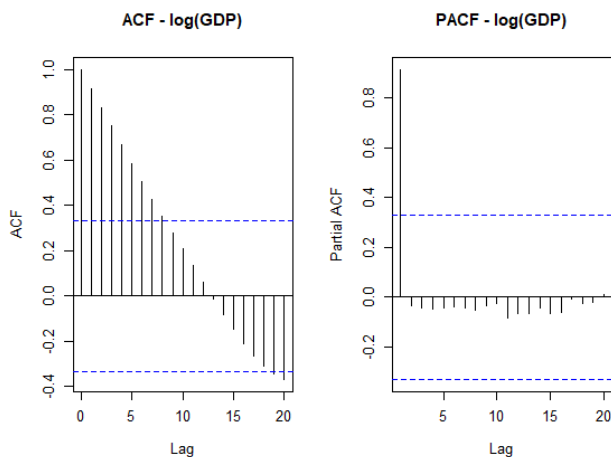
The log-transformed series also remains non-stationary according to the reported tests. The ADF test does not reject a unit root, and the KPSS test rejects stationarity. Consequently, the logarithm alone is insufficient to produce a stationary process.

After first differencing the log GDP series, the evidence becomes mixed. The KPSS test no longer rejects stationarity at the 5% level, whereas the ADF test still fails to reject its unit-root null. This divergence should not be concealed. It means that the formal tests do not provide unanimous confirmation of stationarity after one difference. In the present study, first differencing is nevertheless retained for the candidate ARIMA models because the transformed series is substantially more stable visually and the model-selection and residual diagnostics are conducted on the resulting ARIMA($p, 1, q$) specifications.

Table 1. Stationarity tests for Nepal's nominal GDP series

Test	Statistic	<i>p</i> -value	Interpretation at 5%
ADF (GDP level)	-0.1236	0.9900	Fails to reject unit root
KPSS (GDP level)	0.8527	0.0100	Rejects stationarity
ADF (log GDP)	-2.8638	0.2377	Fails to reject unit root
KPSS (log GDP)	0.9792	0.0100	Rejects stationarity
ADF (first difference of log GDP)	-1.7889	0.6548	Fails to reject unit root
KPSS (first difference of log GDP)	0.1105	0.1000	Does not reject stationarity

This choice should be interpreted as an empirical modeling decision rather than as a claim that every stationarity test gives the same conclusion.

**Figure 2.** First-differenced log GDP**Figure 3.** log GDP ACF-PACF diagnostics.

The ACF and PACF plots are used as identification aids rather than as the sole determinants of model order. With only 35 annual observations, over parameterization can quickly consume degrees of freedom. Candidate models are therefore evaluated jointly using information criteria, residual diagnostics, and parsimony.

Comparison of candidate ARIMA models

Table 2 summarizes the results for the six pre-specified ARIMA candidates.

The model comparison shows that no single candidate dominates every statistic. ARIMA(1,1,2) produces the lowest AIC and AICc among the six pre-specified models, while ARIMA(6,1,2) produces the smallest RMSE and MAD and the largest residual Ljung-Box *p*-value among the candidates shown. The latter model, however, is substantially more parameterized and has less favorable AICc and BIC values. This trade-off illustrates why model selection should not be based on in-sample error alone.

ARIMA(1,1,1), ARIMA(2,1,1), and ARIMA(2,1,2) all have Ljung-Box *p*-values below 0.05 in the comparison table, suggesting remaining residual autocorrelation under the particular test settings used. By contrast, ARIMA(1,1,2), ARIMA(6,1,1), and ARIMA(6,1,2) have *p*-values above 0.05. Among these diagnostically acceptable candidates, ARIMA(1,1,2) offers the strongest AIC and AICc performance and uses fewer parameters than the two high-order alternatives.

For these reasons, ARIMA(1,1,2) is retained as the preferred model *within the restricted candidate set*. It does not mean that ARIMA(1,1,2) is necessarily the globally optimal model among all possible ARIMA specifications.

Automated model-selection robustness check

The automated search produces the following result:

ARIMA(0, 1, 0) with drift,

with estimated drift 0.1146 and standard error 0.0085. The model reports

$AIC = -103.98$, $AICc = -103.59$, $BIC = -100.92$.

These information criteria are lower than those reported for ARIMA(1,1,2). The automated result therefore provides strong evidence in favor of a more parsimonious specification. In economic terms, ARIMA(0,1,0) with drift corresponds to a random-walk-type process in which the differenced log GDP series has a positive average increment.

This finding has two implications. First, the restricted candidate list did not include the automated procedure's preferred simple specification. Second, the empirical evidence does not justify presenting ARIMA(1,1,2) as the only defensible forecasting model. A rigorous interpretation is that ARIMA(1,1,2) performs best among the manually pre-selected candidates, whereas ARIMA(0,1,0)

Table 2. Comparison of pre-specified ARIMA models fitted to log GDP

Model	AIC	AICc	BIC	LogLik	RMSE	MAD	Q-stat	p-value
ARIMA(1,1,1)	-97.0311	-96.2311	-92.4520	51.5155	0.0507	0.0390	22.7112	0.0038
ARIMA(1,1,2)	-98.7618	-97.3825	-92.6563	53.3809	0.0484	0.0373	11.8279	0.1064
ARIMA(2,1,1)	-96.0920	-94.7127	-89.9866	52.0460	0.0498	0.0379	18.5984	0.0095
ARIMA(2,1,2)	-93.8271	-91.6843	-86.1953	51.9136	0.0501	0.0380	19.3147	0.0037
ARIMA(6,1,1)	-95.0360	-89.2760	-82.8251	55.5180	0.0442	0.0318	7.2785	0.0635
ARIMA(6,1,2)	-97.1465	-89.6465	-83.4093	57.5733	0.0411	0.0305	3.6588	0.1605

with drift is the preferred automated information criterion benchmark.

The coexistence of these results is itself informative. Macroeconomic time series can sometimes be forecast competitively by simple persistence plus drift models, especially when long-run trends dominate short-run cyclical structure. Future research should therefore compare forecasts from the two models using an explicit holdout or rolling origin evaluation rather than relying exclusively on in-sample criteria.

Parameter estimates of ARIMA(1,1,2)

The estimated parameters for the selected restricted-candidate model are shown in Table 3.

Table 3. Parameter estimates for ARIMA(1,1,2)

Parameter	Estimate	Std. Error	z-value	p-value
AR(1)	0.999999			
MA(1)	-0.798887	0.179635	-4.4473	8.696×10^{-6}
MA(2)	0.404909	0.247498	1.6360	0.1018

The MA(1) coefficient is strongly different from zero according to the reported asymptotic z -test, whereas the MA(2) coefficient is not statistically significant at the conventional 5% level. More importantly, the AR(1) estimate is essentially equal to one and its standard error is unavailable because the fitted parameter is at or extremely close to the boundary of the parameter space. The R output consequently reports NaN for the standard error and associated test statistics.

This numerical feature is a material limitation of the ARIMA(1,1,2) specification. A boundary estimate can indicate weak identification or near-cancellation between the integrated and autoregressive components. The result does not automatically invalidate the model's fitted values or residual diagnostics, but it weakens the case for treating the estimated coefficient structure as stable and precisely measured. This issue provides an additional reason to report the ARIMA(0,1,0) with drift benchmark prominently.

The estimated innovation variance for ARIMA(1,1,2) is 0.002641, and the log-likelihood is 53.38. The in-sample training RMSE is 0.04837 on the log scale, while the MAE is 0.03735. These values summarize fit within the estimation sample,

Residual diagnostics

The residual analysis provides a second assessment of model adequacy. The Shapiro–Wilk test reports

$$W = 0.95294, \quad p = 0.1395.$$

Thus, at the 5% significance level, the normality null is not rejected. This result does not prove that the residual distribution is exactly normal; rather, it indicates that the available sample does not provide strong evidence against normality.

The `checkresiduals()` Ljung–Box diagnostic reports

$$Q^* = 6.7952, \quad df = 4, \quad p = 0.1471,$$

using a total of seven lags and model degrees of freedom equal to three. Since the p -value exceeds 0.05, there is no statistically significant evidence of residual autocorrelation under this diagnostic configuration.

The residual plot should be interpreted jointly with the formal tests. Visually, the residuals fluctuate around zero without a persistent trend, while the ACF does not show an obvious sequence of dominant autocorrelations. The histogram and Q–Q diagnostic can be used to assess whether the Shapiro–Wilk result is consistent with the observed distribution. Taken together, the diagnostics support the conclusion that the fitted ARIMA(1,1,2) model captures a substantial part of the serial dependence in the differenced log GDP series. Nevertheless, the boundary AR estimate remains a separate parameter-estimation concern.

Five-year GDP forecasts

Table 4 reports forecasts generated from the ARIMA(1,1,2) model and transformed back to the original GDP scale.

The point forecasts imply continued growth in nominal GDP throughout the five-year projection period. The forecast rises from approximately NPR 6.353 trillion in 2025/26 to approximately NPR 7.908 trillion in 2029/30. Because the model is fitted to nominal GDP, these increases can not be interpreted directly as forecasts of real economic growth. Future inflation and price-level movements are embedded in the projected current price aggregate.

The uncertainty surrounding the projections increases rapidly with the forecast horizon. For 2025/26, the 90% prediction interval ranges from NPR 5.838 trillion to NPR 6.913 trillion. By 2029/30, the 90% interval expands from NPR 5.386 trillion to NPR 11.611 trillion. The wide interval at longer horizons demonstrates why

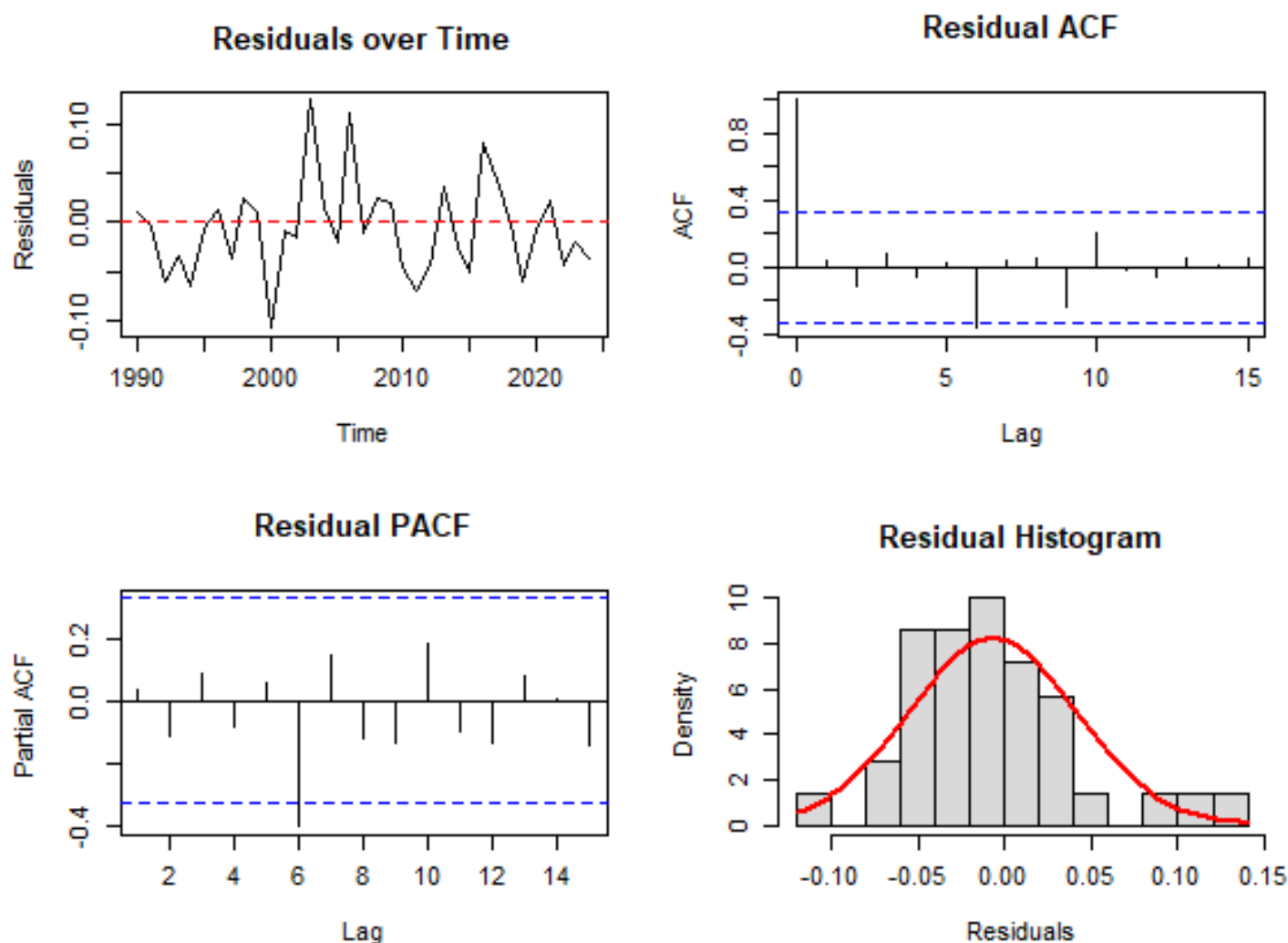


Figure 4. Residual diagnostics for the ARIMA(1,1,2) model.

Table 4. Forecasts of Nepal's nominal GDP, 2025/26–2029/30 (million NPR)

Fiscal year	Forecast	Lower 80%	Upper 80%	Lower 90%	Upper 90%
2025/26	6,353,083	5,948,129	6,785,608	5,838,099	6,913,495
2026/27	6,710,434	6,054,036	7,438,001	5,879,923	7,658,252
2027/28	7,087,885	6,055,846	8,295,805	5,791,631	8,674,261
2028/29	7,486,567	5,986,150	9,363,059	5,618,381	9,975,949
2029/30	7,907,673	5,862,380	10,666,536	5,385,524	11,610,994

point forecasts alone can be misleading. The central projection suggests continued expansion, but the range of statistically plausible outcomes under the model assumptions becomes increasingly broad.

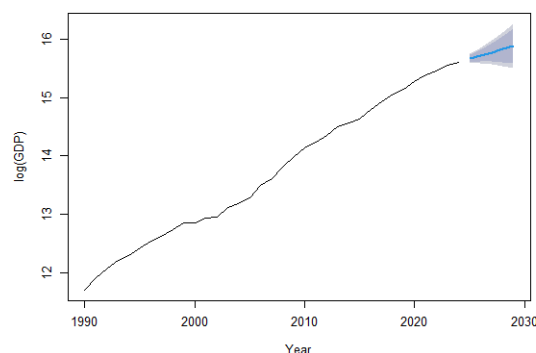


Figure 5. Five-year forecast of Nepal's nominal GDP from ARIMA(1,1,2).

The automated search favors ARIMA(0,1,0) with drift, with substantially lower AIC, AICc, and BIC than the manually selected ARIMA(1,1,2). Moreover, the ARIMA(1,1,2) model has an AR coefficient effectively equal to one with an unavailable standard error. Therefore, this study does not claim that the more complicated model is unambiguously superior.

The findings are broadly consistent with recent forecasting literature in one important respect: simple linear models remain useful benchmarks, but alternative approaches can sometimes improve accuracy. The Nepal Rastra Bank hybrid ARIMA ANN study is particularly relevant because it suggests that Nepal's GDP may contain nonlinear components not fully captured by a purely linear ARIMA structure (Chaudhary and Uprety, 2023). Thus, the present model should be viewed as a transparent baseline and a contribution to reproducible national macroeconomic forecasting rather than as a universal forecasting solution.

A further methodological issue concerns validation. The current results emphasize in-sample information criteria and residual diagnostics. These are necessary for model development but do not replace genuine out-of-sample forecast evaluation. A future extension should reserve one or more recent observations for testing, or use rolling-origin cross-validation, and compare ARIMA(1,1,2), ARIMA(0,1,0) with drift, and alternative methods using common forecast accuracy measures such as RMSE, MAE, MASE, and scale-appropriate percentage errors.

The policy implication is not that the forecast path should be treated as a precise prediction of Nepal's future economy. Instead, the projections provide a quantitative baseline for planning. Ministries, central-bank analysts, development agencies, and researchers can use such a baseline when constructing alternative fiscal, revenue, expenditure, or macroeconomic scenarios.

The results also support the use of parsimonious benchmark models in forecasting exercises. Since the automated ARIMA(0,1,0) with drift model outperforms the restricted candidate models on the reported information criteria, future work should compare this simple benchmark explicitly with higher-order ARIMA models. More complex models should be retained only when they demonstrate clear improvements in out-of-sample accuracy or other relevant forecasting objectives.

Finally, GDP is influenced by factors that are not contained in a univariate historical series. Remittances, tourism, agricultural conditions, public investment, credit conditions, external demand, inflation, exchange rates, and extraordinary shocks may all affect future macroeconomic outcomes. Extensions using ARIMAX, VAR/VECM frameworks, dynamic factor models, machine learning, or hybrid methods may therefore complement the univariate approach.

CONCLUSION

The GDP level and log GDP series show strong evidence of non-stationarity. After first differencing the log series, the KPSS result is consistent with stationarity, whereas

the ADF test continues to fail to reject a unit root. The evidence is therefore mixed and is reported transparently.

Among the six pre-specified ARIMA candidates, ARIMA(1,1,2) has the lowest AIC and AICc and passes the reported residual autocorrelation diagnostic at the 5% level. The residual normality test also does not reject normality. However, ARIMA(6,1,2) has smaller in-sample RMSE and MAD, illustrating that different selection criteria can favor different models. More importantly, the ARIMA(1,1,2) autoregressive coefficient is estimated at a boundary value and has an unavailable standard error.

The automated model-selection procedure identifies ARIMA(0,1,0) with drift and produces substantially lower AIC, AICc, and BIC values than the restricted-candidate winner. Consequently, the most defensible interpretation is not that ARIMA(1,1,2) is universally optimal. Rather, ARIMA(1,1,2) is the preferred specification within the manually selected candidate set, while ARIMA(0,1,0) with drift provides a more parsimonious and information-criterion-favored benchmark.

Forecasts from ARIMA(1,1,2) indicate a continued increase in nominal GDP from NPR 6.353 trillion in 2025/26 to NPR 7.908 trillion in 2029/30. The prediction intervals, however, widen markedly over time, emphasizing that the projections should be used as conditional planning baselines rather than precise future outcomes.

Future research should extend the analysis in three directions: first, through rolling or holdout out-of-sample forecast evaluation; second, through explicit comparison with the random-walk-with-drift benchmark; and third, through multivariate, machine-learning, and hybrid approaches that can incorporate nonlinearities and additional macroeconomic information. Such comparisons would provide stronger evidence about whether additional model complexity produces meaningful forecasting gains for Nepal's economy.

DATA AVAILABILITY

The dataset used for this study covers annual nominal GDP of Nepal from fiscal year 1990/91 to 2024/25. The cleaned study dataset is available from the author's online repository:

https://raw.githubusercontent.com/bijayprad/MPHil/refs/heads/main/data/National%20Data/GDP1990_2025.csv

CONFLICT OF INTEREST

The authors declare no conflict of interest.

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